

MindMatrix

VTU Internship Programme

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Karunada-Kala Source (National Pride)

Android App Development using GenAI · Project Title : Karunada-Kala

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§1 Introduction

1.1 Purpose of this Document

This Product Requirements Document (PRD) defines the full requirements, scope, architecture, and success criteria for Karunada-Kala Source — an Android application developed as part of the MindMatrix VTU Internship Programme (Project Title :90). It serves as the single source of truth for all design, development, testing, and evaluation decisions throughout the internship period.

1.2 Project Background

Traditional Karnataka art forms such as Yakshagana, Kinnala toys, Bidriware craft, Ilkal weaving, and Dollu Kunitha dance are struggling to reach modern audiences. Artisans and performers operating in rural districts lack a digital presence, while culturally curious citizens in cities cannot locate authentic workshops, training centres, or live performance events. Karunada-Kala Source bridges this gap by creating a culturally rich, map-driven Android application that serves as a “Directory of Pride” for Karnataka’s living heritage.

1.3 Intended Audience

- Student Developer(s) building the application
- MindMatrix Mentor — for sprint reviews and deliverable acceptance
- VTU Programme Evaluator — for final grading
- End Users — citizens of Karnataka interested in cultural discovery

1.4 Document Scope

This PRD covers all functional requirements, non-functional requirements, user personas, system architecture, screen specifications, data models, evaluation criteria, and delivery milestones for the Karunada-Kala Source Android application.

2.1 The Core Problem

Karnataka is home to over 50 recognised traditional art forms, yet the majority of citizens — especially the urban youth — cannot name more than two or three. The barriers are not a lack of interest but a lack of discovery infrastructure:

- No centralised digital platform maps artisans, workshops, and performances together.
- Artisan contact information is scattered across social media or entirely offline.
- Tourists and learners cannot find Guru-Shishya workshops without personal connections.
- Performance event information is not aggregated or searchable by location.

2.2 Impact of the Problem

Affected Group	Current Pain	Consequence
Rural Artisans	No digital visibility or direct customer reach	Income loss; craft dies with generation
Urban Learners	Cannot find authentic workshops or teachers	Interest converts to inaction
Cultural Tourists	No event calendar for folk performances	Miss authentic local experiences
Karnataka Govt / NGOs	Difficult to promote heritage digitally	Low ROI on cultural preservation spend

§3 Project Objectives

3.1 Primary Objectives

- Build a fully functional Android application that maps Karnataka artisans and cultural performances using Google Maps SDK.
- Create a rich Art Form Explorer catalogue covering at least 5 traditional art forms with history, images, and AI-generated cultural descriptions.
- Enable direct artisan-to-user connection via a one-tap 'Tap to Call' feature on every Artisan Profile.
- Integrate GenAI (Gemini API) to auto-generate cultural descriptions for each art form, satisfying the programme's GenAI requirement.
- Deliver a polished UI using Karnataka's cultural colour identity (yellow and red from the state flag).

3.2 Secondary Objectives

- Demonstrate clean MVVM architecture with Firebase Firestore as the cloud data store.
- Achieve all 6 Success Criteria defined in the original project brief (§16).
- Produce all required internship deliverables: wireframe, prototype, APK, project report, and video demo.

3.3 GenAI Integration Objective

The application must visibly leverage a Generative AI API (Gemini API recommended) to generate cultural content. This must be demonstrable live to the evaluator on the Art Form Detail screen. The AI-generated text must be clearly labelled as 'AI-generated' within the UI.

Attribute	Persona A — Arjun (Learner)	Persona B — Savitha (Artisan)
Age / Location	23, Bengaluru — software intern	47, Dharwad — Yakshagana artist
Goal	Find a Kuchipudi workshop near him; learn about art forms he sees on social media	Get discovered by urban learners and tourists; receive direct booking calls
Frustration	Google shows commercial dance schools, not traditional artisans	Has no website; relies only on word-of-mouth referrals
App Use	Browses Art Form Explorer → finds artisan on map → calls to enquire about workshop	Profile listed by admin; receives calls directly from app users
Device	Android 12, mid-range phone, 4G	Android 9, entry-level phone, spotty connectivity

§5 Project Scope

5.1 In-Scope Features

ID	Feature	Description	Priority	Week
F-01	Art Form Explorer	LazyVerticalGrid of Karnataka art forms. Each card shows name, district, and thumbnail. Tap → Detail screen.	Must Have	1
F-02	Art Form Detail + GenAI	Full-screen detail with history, high-res image (Glide), and a 'Generate AI Description' button that calls Gemini API and renders the response.	Must Have	1
F-03	Artisan Map	Google Maps screen. Two custom marker types: Workshop (blue) and Performance (red star). Tap marker → bottom sheet with artisan preview.	Must Have	1
F-04	Artisan Profile	Dedicated screen with photo, name, district, art form, short bio, and prominent 'Tap to Call' button (Intent.ACTION_DIAL).	Must Have	1-2
F-05	Event Feed	Chronological list of upcoming Yakshagana / Dollu Kunitha performances. Data seeded in Firestore. Shown on Home screen.	Must Have	2
F-06	Workshop Sign-up Form	Simple form (name, phone, art form interest) that logs interest to Firestore. No backend validation required for internship scope.	Should Have	2
F-07	Search / Filter	Search bar on Art Form Explorer filtered by art form name or district.	Could Have	2

5.2 Out-of-Scope Exclusions

Excluded Feature	Rationale	Future Scope (Phase 2)
In-app video streaming	Requires CDN infrastructure beyond internship scope	Integrate ExoPlayer + YouTube Data API in Phase 2
User login / authentication	Firebase Auth adds complexity; not required for MVP demo	Add Google Sign-In for personalised bookmarking
Payment / booking transactions	Requires payment gateway compliance (PCI-DSS)	Integrate Razorpay for workshop seat booking
Artisan self-registration portal	Admin seeding of data is sufficient for demo	Build artisan-facing registration flow in Phase 2
Offline-first architecture	Firebase online mode adequate for demo; offline adds complexity	Add Room DB mirror of Firestore for full offline support

6.1 Art Form Explorer (F-01)

- FR-01: The system shall display all art forms stored in Firestore as cards in a 2-column LazyVerticalGrid.
- FR-02: Each card shall display the art form name, home district, and a thumbnail image loaded via Glide.
- FR-03: Tapping a card shall navigate to the Art Form Detail screen (F-02) passing the art form ID.

6.2 Art Form Detail + GenAI (F-02)

- FR-04: The detail screen shall display the art form name, full history text, and a high-resolution image.
- FR-05: The screen shall include a 'Generate AI Description' button visible below the history text.
- FR-06: On button tap, the app shall call the Gemini API with a prompt referencing the art form name and return a 3-sentence cultural description rendered in a styled card labelled 'AI-Generated'.
- FR-07: A loading indicator shall be displayed while the API call is in progress.
- FR-08: If the API call fails, a user-friendly error message shall be shown (not a raw exception).

6.3 Artisan Map (F-03)

- FR-09: The map screen shall load a Google Map centred on Karnataka (lat 15.3, lng 75.7, zoom 7).
- FR-10: Each artisan in Firestore shall be rendered as a custom marker. Workshop artisans use a blue marker; Performance artisans use a red star marker.
- FR-11: Tapping a marker shall display a bottom sheet showing the artisan's name, art form, and district, with a 'View Profile' button.
- FR-12: Tapping 'View Profile' shall navigate to the Artisan Profile screen (F-04).

6.4 Artisan Profile (F-04)

- FR-13: The profile screen shall display artisan photo (Glide), full name, district, art form speciality, and a 2–3 sentence bio.
- FR-14: The screen shall include a prominent 'Tap to Call' button that triggers Intent.ACTION_DIAL with the artisan's phone number pre-filled.
- FR-15: The artisan type (Workshop / Performance) shall be displayed as a coloured badge on the profile.

6.5 Event Feed (F-05)

- FR-16: The Home screen shall include a horizontal-scrolling event feed showing at least 5 upcoming events from Firestore.
- FR-17: Each event card shall display the event title, art form, date, and venue district.
- FR-18: Events shall be sorted by date ascending (nearest first).

§7 Non-Functional Requirements

ID	Category	Requirement	Test Method	Pass Threshold
NFR-01	Performance	Art Form Explorer screen shall load all cards within 3 seconds on a 4G connection.	Android Profiler, 3 runs avg	< 3,000 ms
NFR-02	Performance	Google Maps screen shall render all markers within 4 seconds of screen open.	Stopwatch + Logcat	< 4,000 ms
NFR-03	Compatibility	App shall run without crashes on Android 9.0 (API 28) and above.	API 28 emulator full regression	Zero crashes
NFR-04	Usability	Any primary user task (find artisan, tap to call) shall be completable in 3 taps or fewer from the Home screen.	Manual task walkthrough	≤ 3 taps
NFR-05	UI / Brand	All screens shall use Karnataka flag colours (yellow #FDD835 and red #C62828) as accent colours consistently.	Visual design review vs spec	100% screens compliant
NFR-06	Code Quality	Architecture shall follow MVVM. Zero Firestore/API calls shall appear in Composable functions directly.	Code review grep	0 violations
NFR-07	Reliability	App shall not crash on back-navigation from any screen.	Manual back-stack testing	Zero crashes

8.1 Architecture Pattern

The application shall follow the MVVM (Model-View-ViewModel) architecture pattern as mandated by the MindMatrix programme standards. The three layers are strictly separated:

- View Layer: Jetpack Compose Composable functions. Observes StateFlow from ViewModel. Zero business logic.
- ViewModel Layer: Holds UI state as StateFlow/LiveData. Calls Repository functions. Survives configuration changes.
- Model / Repository Layer: Abstracts all data sources (Firestore, Gemini API). ViewModels never call Firestore directly.

8.2 Technology Stack

Component	Technology	Purpose
Language	Kotlin	Primary development language
UI Framework	Jetpack Compose	All screens and UI components
Architecture	MVVM + Repository	Separation of concerns; testability
Cloud Database	Firebase Firestore	Art form data, artisan profiles, events, sign-ups
Maps	Google Maps SDK for Android	Artisan location display with custom markers
Image Loading	Glide	Async image loading and caching for artisan photos
GenAI	Gemini API (Google AI SDK)	Cultural description generation on Art Form Detail

Navigation	Jetpack Navigation Compose	Screen routing and back-stack management
Async	Kotlin Coroutines + Flow	All network and Firestore operations
DI	Manual DI (or Hilt optional)	Repository injection into ViewModels
Min SDK	API 28 (Android 9.0)	NFR-03 compatibility requirement

8.3 Firestore Data Model

Collection: artforms

Field	Type	Description / Example
id	String	"yakshagana"
name	String	"Yakshagana"
district	String	"Uttara Kannada"
history	String	Long-form history text (200–300 words)
imageUrl	String	Firebase Storage URL for high-res image
thumbnailUrl	String	Compressed thumbnail for Explorer grid

Collection: artisans

Field	Type	Description / Example
id	String	"artisan_001"
name	String	"Ramesh Gowda"
artFormId	String	Reference to artforms collection
district	String	"Mysuru"
type	String	"workshop" or "performance"
phone	String	" +91XXXXXXXXXX"
bio	String	2–3 sentence artisan biography
photoUrl	String	Firebase Storage URL
latitude	Double	GPS latitude for map pin
longitude	Double	GPS longitude for map pin

Collection: events

Field	Type	Description / Example
id	String	"event_001"
title	String	"Yakshagana Utsav 2026"
artFormId	String	Reference to artforms
date	Timestamp	Firebase Timestamp, sorted ascending
venue	String	"Town Hall, Dharwad"
district	String	"Dharwad"

§9 User Flow

9.1 Primary Navigation Flow

Home Screen → Bottom Navigation Bar → 3 tabs: [Explore] [Map] [Events]

9.2 Art Form Discovery Flow

- User opens app → Home Screen shows Event Feed + 'Explore Arts' shortcut
- Tap 'Explore' → Art Form Explorer (F-01) grid loads
- Tap art form card → Art Form Detail screen (F-02)
- Tap 'Generate AI Description' → Gemini API called → description renders

9.3 Artisan Discovery Flow

- Tap 'Map' tab → Artisan Map (F-03) loads with markers
- Tap marker → Bottom sheet shows artisan preview
- Tap 'View Profile' → Artisan Profile screen (F-04)
- Tap 'Tap to Call' → Phone dialler opens with number pre-filled

9.4 Workshop Registration Flow

- From Artisan Profile → Tap 'Register for Workshop' → Sign-up form (F-06)
- User fills name + phone + art form interest → Tap Submit
- Data written to Firestore 'signups' collection → Success confirmation shown

ID	Screen Name	Key UI Elements	Colour / Style Notes
S-01	Home Screen	App logo, tagline, horizontal event feed cards, Explore shortcut button	Dark indigo toolbar; yellow accent CTA button
S-02	Art Form Explorer	Search bar, 2-col LazyVerticalGrid, art form cards with thumbnail + name + district badge	White background; district badge in red #C62828
S-03	Art Form Detail	Hero image (full-width), name, history text, 'Generate AI Description' button, AI result card labelled 'Powered by Gemini'	Yellow #FDD835 button; AI card with light purple bg
S-04	Artisan Map	Full-screen Google Map, custom blue/red markers, bottom sheet on tap with artisan preview	Marker colours: blue = workshop, red = performance
S-05	Artisan Profile	Circular photo, name, art form badge, district, bio text, large 'Tap to Call' FAB button	Red #C62828 FAB; yellow accent underlines
S-06	Workshop Sign-up	Name field, phone field, art form dropdown, Submit button, success screen	Clean white form; indigo Submit button

§11 GenAI Integration Specification

11.1 API Used

Google Gemini API via the Google AI Android SDK (com.google.ai.client.generativeai). Model: gemini-1.5-flash (cost-efficient, fast responses suitable for demo).

11.2 Prompt Template

"Write a 3-sentence cultural description of {artFormName} from Karnataka, India, suitable for a heritage discovery app. Include its historical origin, what makes it unique, and why it matters for Karnataka's identity. Write in simple, engaging English."

11.3 Implementation Notes

- API key stored in local.properties (never committed to version control).
- Call made in ViewModel using viewModelScope.launch coroutine.
- UI state: Idle → Loading (spinner visible) → Success (text rendered) → Error (friendly message).
- Generated text cached in ViewModel for the session (no repeated API calls on recomposition).
- AI result card must include the label 'AI-Generated · Powered by Gemini' in small italic text below the description.

M#	Day	Milestone Description	Evidence / Demo	Status
M1	Day 1	Project setup — Android Studio project, Firestore connected, folder structure (MVVM), colour theme applied	App launches on emulator. Firestore read confirmed in Logcat.	☐ Not Started
M2	Day 3	Art Form Explorer + Detail screens complete with Firestore data and Glide images	Live demo: grid loads, tap navigates to detail, image renders.	☐ Not Started
M3	Day 5	Gemini API integrated — 'Generate' button visible, API call works, result renders on screen	Live demo: tap button → spinner → AI text appears.	☐ Not Started
M4	Day 7	Google Maps screen complete with custom markers for all seeded artisans	Live demo: map loads, both marker types visible, tap shows bottom sheet.	☐ Not Started
M5	Day 9	Artisan Profile + Tap to Call working. Event Feed on Home screen complete.	Live demo: tap call button → dialler opens with number.	☐ Not Started
M6	Day 11	Workshop Sign-up form complete. All screens UI-polished with Karnataka colours.	Side-by-side: spec colours vs built screens.	☐ Not Started
M7	Day 13	Full regression testing on API 28 emulator. All NFRs tested.	NFR test results table filled. Zero crashes recorded.	☐ Not Started
M8	Day 14	All deliverables submitted: APK, project report, video demo, wireframe.	Mentor formal acceptance. Closeout meeting completed.	☐ Not Started

§13 Deliverables

ID	Deliverable	Owner	Due	Acceptance Condition
D-01	Figma Wireframes — all 6 screens with Karnataka colour palette	[UI Developer]	Day 2	Mentor written approval. All S-01 to S-06 covered.
D-02	Firestore Schema Diagram — 3 collections with fields	[Firebase Handler]	Day 2	Diagram matches §8.3 data model. Mentor acknowledged.
D-03	Functional UI Prototype — all screens navigable with mock data	[UI Developer]	Day 4	Live demo: all 6 screens reachable, no crashes.
D-04	Final Working APK — signed, tested on API 28	[Feature Integrator]	Day 13	All Must Have FRs pass. NFR-03 zero crashes on API 28.
D-05	Project Report — architecture, screenshots, PRD section refs	[Team Lead]	Day 14	All required sections present. Features vs spec table included.
D-06	Video Demo — 2–3 minutes narrated, all features shown live	[Team Lead]	Day 14	All F-01 to F-05 demonstrated. GenAI visibly triggered. Timing 2–3 min.

14.1 Art Forms (minimum 5 required)

ID	Name	District	Notes for Seed
yakshagana	Yakshagana	Uttara Kannada	Night-long musical theatre with elaborate costumes and headgear
bidriware	Bidriware	Bidar	Metal craft with silver inlay on blackened zinc-copper alloy
kinnala	Kinnala Toys	Koppal	Brightly painted wooden toys; GI-tagged craft
ilkal	Ilkal Saree Weaving	Bagalkot	Hand-woven silk-cotton sarees with distinctive tope teni border
dollu	Dollu Kunittha	Tumkur	High-energy drum dance performed by the Kuruba community

14.2 Artisans (minimum 6 required — 3 workshop, 3 performance)

Name	Art Form	District	Type	Approx Coordinates
Ramesh Gowda	Yakshagana	Dharwad	Performance	lat: 15.46, lng: 75.01
Kaveri Devi	Bidriware	Bidar	Workshop	lat: 17.92, lng: 77.52
Suresh Chitari	Kinnala Toys	Koppal	Workshop	lat: 15.35, lng: 76.15
Annapurna Bai	Ilkal Saree	Bagalkot	Workshop	lat: 16.18, lng: 75.69
Manjunath Tala	Dollu Kunittha	Tumkur	Performance	lat: 13.34, lng: 77.10
Girija Naik	Yakshagana	Udupi	Performance	lat: 13.34, lng: 74.74

§15 Future Scope (Phase 2 Recommendations)

Feature	Description	Technology
In-app video player	Embed Yakshagana / Dollu Kunita performance clips directly	ExoPlayer + YouTube Data API v3
Artisan self-registration	Let artisans create and manage their own profiles	Firebase Auth + Firestore write rules
Workshop booking & payment	Reserve seats and pay for workshops in-app	Razorpay SDK
Multilingual support	Full Kannada language UI option for rural users	Android strings.xml localisation
Offline mode	Cache art form data locally for no-connectivity areas	Room DB mirroring Firestore
AI chatbot guide	Conversational assistant to recommend art forms based on interest	Gemini API multi-turn conversation

16.1 Evaluation Weightage

Criterion	Weight	Acceptance Standard	How to Test
Functionality	30%	All F-01 to F-05 Must Have features work end-to-end with live Firebase data. Zero crashes on primary flows.	Execute all FR test cases. Binary pass/fail per feature.
UI / UX Design	25%	All screens match §10 colour spec. Navigation correct. Karnataka flag colours used consistently.	Side-by-side: §10 spec vs built screens.
Code Quality	20%	MVVM per §8: zero Firestore calls in Composable files. All state as StateFlow.	Code review grep for Firestore in screen files.
Innovation (GenAI)	15%	Gemini API visibly integrated. AI text labelled. Feature demonstrable live in 30 seconds.	Live demo: tap button → AI result visible.
Presentation	10%	Video 2–3 min. Report references PRD § numbers. All D-01 to D-06 submitted.	Report review. Video timing check.

16.2 Original Project Brief — 6 Success Criteria

#	Success Criterion (from brief)	How Met in This App
SC-1	The map must show different icons for 'Workshops' vs 'Performances'	F-03: Blue marker = Workshop; Red star = Performance
SC-2	The 'Artisan Profile' must have a 'Tap to Call' feature	F-04: Intent.ACTION_DIAL with pre-filled number
SC-3	The UI must use the yellow and red colours of the Karnataka flag tastefully	§10: Yellow #FDD835 and red #C62828 used as accent colours across all screens
SC-4	Art Form Explorer must be present with catalog and history	F-01 + F-02: Grid catalog with full detail and history text

SC-5	GenAI must be visibly integrated	F-02: Gemini API generates cultural description live on Art Form Detail screen
SC-6	Event feed for upcoming performances	F-05: Chronological event list on Home screen from Firestore

ID	Risk Description	P	I	Score	Zone	Response	Owner
R-01	Google Maps API key not set up correctly — blank map screen	3	5	15	RED	Test Maps setup on Day 1 before any other feature. Follow official quickstart.	[Dev]
R-02	Gemini API quota exceeded during demo	2	5	10	AMB	Cache last AI response in ViewModel. Use free tier + keep demo calls minimal.	[Dev]
R-03	Firestore rules block reads during evaluator demo	3	4	12	AMB	Set Firestore rules to public read for demo. Document this in report.	[Dev]
R-04	Custom map markers not rendering (resource not found crash)	3	4	12	AMB	Test markers on Day 6 separately. Have fallback to default markers.	[Dev]
R-05	Scope creep — adding video player or auth eats into Week 2	4	4	16	RED	Post §5.2 out-of-scope list visibly in workspace. No new features after Day 8.	[Lead]
R-06	Documentation underestimated — report + video take longer than expected	4	4	16	RED	Take screenshots from Day 1. Start report on Day 10. Script video on Day 12.	[Lead]
R-07	Glide image loading fails on emulator (missing internet permission)	3	3	9	AMB	Add INTERNET permission to AndroidManifest.xml on Day 1. Test images Day 2.	[Dev]

§18 Glossary

Term	Definition
PRD	Product Requirements Document — this document. The single source of truth for all requirements.
MVVM	Model-View-ViewModel — Android architecture separating UI (Composables), logic (ViewModels), and data (Repository).
Firestore	Google Firebase's NoSQL cloud database used to store art form, artisan, and event data.
Gemini API	Google's GenAI API used to generate cultural descriptions for art forms in this app.
GenAI	Generative Artificial Intelligence — AI that generates new content (text, images) from prompts.
Jetpack Compose	Android's modern declarative UI toolkit used to build all screens in this app.
Glide	Android image loading library used to load and cache artisan photos and art form images.
StateFlow	Kotlin coroutine-based observable state holder used in ViewModels to expose UI state.
Guru-Shishya	Traditional Indian teacher-student relationship; the primary model for learning classical art forms.
GI Tag	Geographical Indication tag — a certification that a product originates from a specific region (e.g., Kinnala Toys).
MVP	Minimum Viable Product — the minimum feature set satisfying all Must Have items that passes evaluation.
FR	Functional Requirement — a specific behaviour the system must exhibit. Format: FR-XX.
NFR	Non-Functional Requirement — a quality constraint (performance, compatibility, usability).

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